American Computer Science League

2025-2026 ● Contest 1: Short Problems ● Senior Division

1. Computer Number Systems

Evaluate and express the answer in hexadecimal.

$100_2 \bullet (42_8 + 375_8)$

A. 43700
B. 11F
C. 437
D. 47C
E. 2174

2. Computer Number Systems

In what year after 2025_{10} will the year, expressed in hexadecimal, first be a palindrome?

A. 707
B. 7E7
C. 7F7
D. 8E8
E. 9E9

3. Recursive Functions

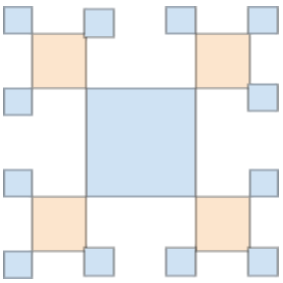
Find $f(f(f(7)))$ given:

$$f(x) = \begin{cases} f(x+3) - 1 & \text{if } x \text{ is a prime number greater than } 5 \\ f(x-1) + 2 & \text{if } x = 1 \text{ or } x \text{ is a multiple of either } 3 \text{ or } 5 \\ x - 2 & \text{otherwise} \end{cases}$$

A. 6
B. 7
C. 8
D. 9
E. 10

4. Recursive Functions

Step 1: Draw a square.
Step 2: Draw new squares at each corner of the original square, each of which is half the length.
Repeat Step 2 four more times without using the connected corner.
How many squares were added during the final repetition?



A. 108
B. 243
C. 256
D. 324
E. 1024

5. What Does This Program Do?

What is the output of the following program if array *arr* is filled with the following values?

arr=	1000	500	100	50	10	5	1

```
letters = "MDCLXVI"
num = 1776
ans = ""
for x = 0 to 6
    val = int(num / arr(x))
    for y = 1 to val
        ans = ans + letters[x]
    next y
    num = num % arr(x)
next x
output ans
```

A. MDCCCLXVI
B. MDCCLXXVI
C. MDCLXXVII
D. MDCCXXXVI
E. DDCCLXXVI

6. What Does This Program Do?

What is the output of the following program if the function str(*n*) makes the number *n* a string value?

```
n = 0
for x = 1 to 4
    z = 10 - x
    a = str(1089 * x)
    s = ""
    for y = 0 to len(a)-1
        s = a[y] + s
    next y
    if s == str(1089 * z)
        n = n + 1
    end if
next x
output n
```

A. 4
B. 3
C. 2
D. 1
E. 0